

Little Red Dots: Stellar Eruptions in Disguise?

New model links LRDs to supernovae and Eta Carinae's Great Eruption.

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Background

JWST's Little Red Dots are compact, red sources with broad Balmer lines and blackbody-like continua. Their nature is debated, with some suggesting overmassive black holes.

LRDs show features not seen together in any other galaxy or AGN, challenging standard models.

What this paper is selling

The authors propose LRDs are super-Eddington, intermediate-mass engines enshrouded in dense winds, mimicking stellar eruptions. This explains their spectra and variability, and suggests lower black hole masses.

Key results to remember

- LRDs cluster at ~5000 K, like Type II_n SNe and Eta Carinae.
- Wind-shock model reproduces line profiles and Balmer breaks.
- Implied engine masses are <10⁵ solar masses, not overmassive.

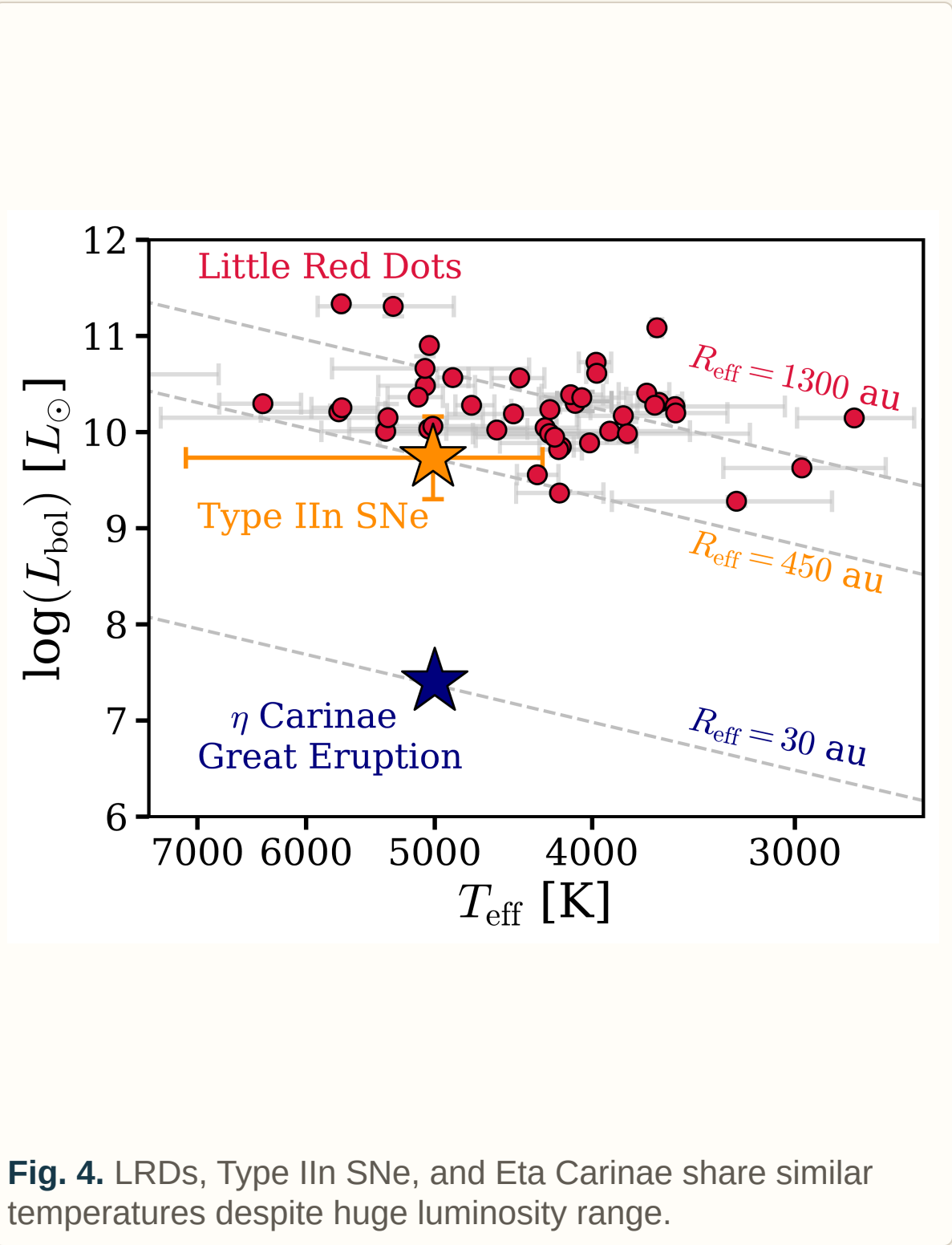


Fig. 4. LRDs, Type II_n SNe, and Eta Carinae share similar temperatures despite huge luminosity range.

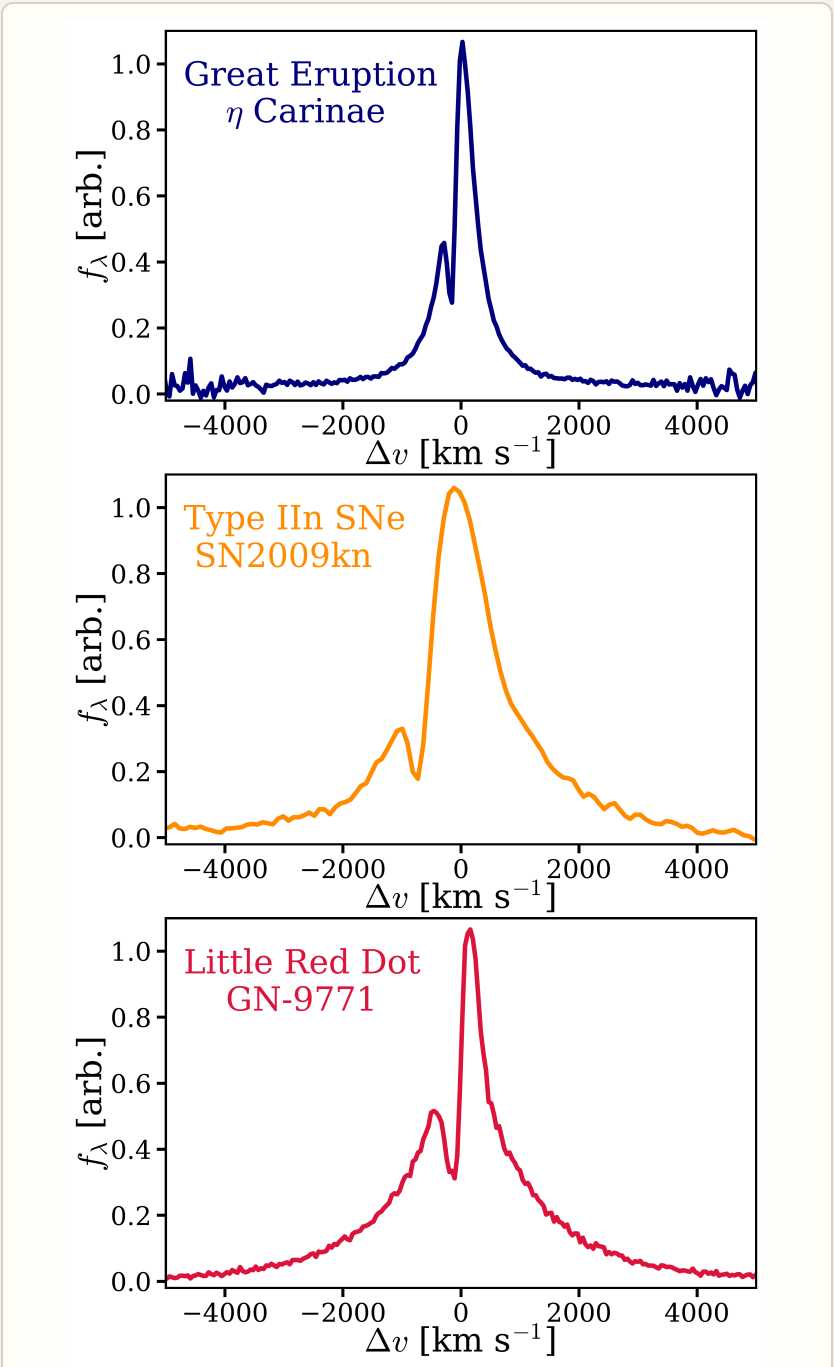


Fig. 1. H α profiles of Eta Carinae, Type II_n SNe, and LRDs show striking similarities.